

- 30 The cathode of a cell is a square piece of copper with dimensions $0.1\text{ m} \times 0.1\text{ m}$. The electrolyte is copper(II) sulfate.

Assume that each copper atom occupies a cube of length $3.0 \times 10^{-12}\text{ m}$, the piece of copper has no thickness and that there is a uniform coverage.

The Faraday constant = $9.65 \times 10^4\text{ C mol}^{-1}$.

How long will it take a current of 4.0 A to cover **both sides** of the piece of copper with new copper to a total depth of 1000 atoms?

- A** 178 s **B** 24.7 h **C** 49.4 h **D** 98.9 h